# D1.2 Protein synthesis

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 45 |

**About 45 minutes,** or one section at a time. You will need a table of mRNA codons. The number of marks shows how many separate points to make.

## Section A: Recall

9 MARKS

1 **STATE** the name of the enzyme that carries out transcription. [D1.2.1] [1]

2 A DNA template strand reads C A T G G A. **WRITE** the mRNA. [D1.2.2] [1]

3 **DISTINGUISH** between a codon and an anticodon. [D1.2.7] [2]

4 **DEFINE** gene expression. [D1.2.4] [1]

5 **STATE** the type of bond that joins amino acids in a polypeptide. [D1.2.10] [1]

6 **STATE** the number of different codons, and the number of amino acids they code for. [D1.2.8] [2]

7 **STATE** the part of a ribosome to which mRNA binds. [D1.2.6] [1]

## Section B: Understanding

16 MARKS

8 **DESCRIBE** the process of transcription. [D1.2.1, D1.2.2] [4]

9 **EXPLAIN** why a triplet code is needed, and not a code of two bases. [D1.2.8] [2]

10 **EXPLAIN** what the universality of the genetic code suggests about living organisms. [D1.2.8] [2]

11 **EXPLAIN** why stability of DNA is important in cells that do not divide. [D1.2.3] [2]

12 Liver cells and neurons in one person contain the same genes. **EXPLAIN** why they make different proteins. [D1.2.4] [2]

13 **DESCRIBE** one cycle of elongation in translation. [D1.2.10] [4]

## Section C: Working with sequences

12 MARKS

14 [D1.2.9]

A section of mRNA reads: A U G  U U U  C G U  G G A  U A A

a **DEDUCE** the amino acid sequence. [2]

b **DEDUCE** the DNA template strand for the first three codons. [1]

c **STATE** the anticodon for the second codon. [1]

d The third codon changes from C G U to C G C. **DEDUCE** the effect on the polypeptide, giving a reason. [2]

15 [D1.2.11]

Part of the non-template strand of a normal gene for beta-globin reads C C T G A G G A G. In the sickle cell allele, it reads C C T G T G G A G.

a DEDUCE the mRNA for each allele. [2]

b DEDUCE how the amino acid sequence differs. [2]

c OUTLINE how this changes red blood cells at low oxygen. [2]

## Section D: Longer answer

8 MARKS

16 EXPLAIN how the information in a gene is used to make a polypeptide. [D1.2.1–10] [8]